Anatomical Science International (2025) 100:141–142
https://doi.org/10.1007/s12565-025-00824-7

EDITORIAL

Featuring the morphology: impossible, inseparable, instable, innumerable and infinite

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Published online: 24 January 2025
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Morphology is a classic branch of biological science that focuses on the shape, structure, and conformation of living organisms. It has a long history and serves as a foundation for the development of natural sciences, particularly in image description and classification. Even genetics in its infancy, as represented by Gregor Johann Mendel (1822–1884), paved the way for classifying phenotypes, which formed the basis for calculating hereditary ratios and establishing the laws of genetics (Mendel 1). Comparative anatomy is one of the most prominent fields where morphology plays a pivotal role in identifying common features to establish structural concepts. However, from a contemporary scientific perspective dominated by quantification and reductionism, morphology sometimes faces prejudice and discrimination. While morphometry, allows us to quantify morphological features, it captures only part of the complex shape in numerical values. Most morphological features cannot be explicitly reproduced using mathematical methods. This approximation can trace the shape; however, recapitulation is impossible.

Paul Weiss (1898–1989), an American biologist, is one of those who highlighted the limitation and disadvantages of applying reductionism to the study of living thing. He is best known for his thought experiment with the chick embryo (Weiss 4). When a vital chick embryo is compared with a whole homogenate of the same embryo, both can be regarded as identical in terms of weight and chemical composition. According to his interpretation, it is the structural organisation that is lost when the embryo is crushed into a homogenate. The loss of structural organisation results in the disintegrity of the embryo, simply put, death. From a morphological standpoint, both the shape and structure are lost by homogenisation. Structure can be defined as a morphological framework that harbours a series of functions that cooperate to maintain the milieu intérieur of living organisms. The shape refers to the external appearance maintained by the orchestration of each molecular component of the whole body. Therefore, crushing the intact embryo by homogenisation leads to the disorganisation of the complex and interdependent functions residing in the structure. We can generalise his thought experiment to the biological sciences and conclude that both shape and function are inseparable.

A completely unexpected aspect of this morphology was its dynamicity. While shape and form may appear unchanging, they constantly undergo transformations due to metabolic activities over time. Therefore, morphology must be regarded as an instantaneous snapshot at a specific time point. When the temporal change in shape occurs slower than the resolution of our consciousness, it seems to remain unchanged, creating an unconsciously coherent visual image. By narrowing the temporal resolution infinite to zero, the shape and form can be fixed and become static, as represented by chemically or physically (frozen) fixed tissues. In this sense, shape and function are separable; however, we cannot affirm that fixed tissues reflect the real shape of the living state. In contrast, let us consider developmental processes as an example to determine the fleetingness of a shape subjected to extinction. Five pharyngeal arches exist in mammals, as described in almost every developmental biology textbook. However, six arches were initially formed, and the 5th was ephemeral. Even though other major pharyngeal arches continue to exist for a relatively longer period than the 5th, they are no more than instable.

Motoo Ōtaguro (1893–1979), a Japanese music critic and photographic artist, expressed his ideas on photography (Ōtaguro 2): “I believe the glamour side of photography is to find a composition of fine arts in an ordinary object all around us. This is the most important attribute for artists, especially photographers, who find the utmost extraordinariness out of ordinariness. Fascinating themes are right next to us and exist as usual in daily life (English interpretation by

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the author, ST).” When applied to morphology, Ōtaguro’s statement can be extrapolated in the following manner: themes of morphological research still reside but are concealed on well-known phenomena or structures, those of which were invisible by non-expertise’s eye. However, having an eye on the morphology and implementation of state-of-the-art technologies makes it possible to recognise, as asserted in the previous Editorial (Takeda 3). In this regard, morphologists are on the same horizon as photographic artists. Similar to photography, morphological objects are innumerable.

Finally, I sincerely appreciate your continued interest in and support for *Anatomical Science International* (*ASI*) during my duties as EIC from 2021 to 2025. *ASI* remains committed to advancing our journey in morphology. We welcome new submissions and look forward to receiving original, thought-provoking contributions to *ASI*. The possibilities are infinite.

References

Mendel GJ (1865) Versuche über Pflanzen-Hybriden. Verhandlungen des naturforschenden Vereines in Brünn. IV-Band. 3–47

Ōtaguro M (1921) Sha Shin Shoh-ron (Short essay on photography). Sha Shin Geijutsu. June. (**In Japanese**)

Takeda S (2023) Whereof one cannot see, thereof one must be silent. Revealing the underlying morphology: challenging reductionism through 3D visualisation for future anatomy. Anat Sci Int 98:307–308

Weiss P (1962) From cell to molecule. 1–72. In: Allen MJ (ed) The molecular control of cellular activity. McGraw-Hill Book Company, Inc., New York

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